IS101 — Excel E4: Formulas & Functions

College of Southern Nevada

Practice Instructions

Download and open your **Excel_E4_Starter** file. Work through each skill below using the steps provided — each skill lists what to do, exactly how to do it, and what the result should look like. Check off each skill on the IS101 reference page as you complete it.

Skills in This Guide

1 Use Relative, Absolute, and Mixed References
2 Use Structured References in Formulas
3 Use AVERAGE, MAX, MIN, and SUM
4 Use COUNT, COUNTA, and COUNTBLANK
5 Use the IF Function
6 Use the SORT() Function
7 Use the UNIQUE() Function
8 Use LEFT, RIGHT, and MID
9 Use UPPER, LOWER, and LEN
10 Use CONCAT and TEXTJOIN
11 Use a Nested IF
12 Use COUNTIF
13 Use SUMIF
14 Use TODAY and NOW
15 Use YEAR, MONTH, and DAY
16 Use VLOOKUP
17 Use XLOOKUP
18 Use IFERROR

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1. Use Relative, Absolute, and Mixed References

Your Task: Write one formula that can be filled down and across without its reference to a fixed tax-rate cell shifting.

STEP-BY-STEP INSTRUCTIONS

1. Click a formula cell and type =B2*D1 (D1 holds the tax rate), then press Enter.
2. Select that cell and drag its fill handle down the column.
3. Press F4 while editing a reference to cycle between relative, absolute, and mixed forms.

WHAT THE RESULT LOOKS LIKE

Every filled cell multiplies its own row's value by the same fixed D1 cell, because the $ signs locked that reference in place.

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2. Use Structured References in Formulas

Your Task: Write a formula that refers to a Table's column by name instead of by cell address.

STEP-BY-STEP INSTRUCTIONS

1. Click a cell inside (or referencing) the Excel Table.
2. Type =SUM(then click a cell in the target column so Excel inserts the structured reference automatically.
3. Press Enter to complete the formula.

WHAT THE RESULT LOOKS LIKE

The formula bar shows something like =SUM(Table1[Amount]) and automatically expands if new rows are added to the table.

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3. Use AVERAGE, MAX, MIN, and SUM

Your Task: Summarize a column of numbers with its total, average, highest, and lowest value.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell and type =SUM(range), press Enter.
2. In the next cell type =AVERAGE(range), press Enter.
3. Then use =MAX(range) and =MIN(range) in two more cells.

WHAT THE RESULT LOOKS LIKE

Each formula returns the corresponding total, average, maximum, or minimum of the selected numbers.

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4. Use COUNT, COUNTA, and COUNTBLANK

Your Task: Count how many numeric entries, how many non-empty entries, and how many empty cells a column has.

STEP-BY-STEP INSTRUCTIONS

1. Type =COUNT(range) in a blank cell for numeric-only cells.
2. Type =COUNTA(range) in another cell for all non-empty cells.
3. Type =COUNTBLANK(range) in a third cell for empty cells.

WHAT THE RESULT LOOKS LIKE

The three formulas return, respectively, the count of numbers, the count of any filled cell, and the count of empty cells in that range.

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5. Use the IF Function

Your Task: Mark each row Pass or Fail depending on whether its score meets a passing threshold.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell next to the first score.
2. Type =IF(A2>=70,"Pass","Fail") and press Enter.
3. Fill the formula down for the remaining rows.

WHAT THE RESULT LOOKS LIKE

Each row now shows Pass or Fail based on its own score, updating automatically if a score changes.

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6. Use the SORT() Function

Your Task: Produce a sorted copy of a data range without disturbing the original data's order.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell to the side of the data.
2. Type =SORT(range,1,-1) to sort descending by the first column, then press Enter.

WHAT THE RESULT LOOKS LIKE

A sorted copy of the range spills out from the formula cell automatically; the original range is untouched.

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7. Use the UNIQUE() Function

Your Task: List each distinct value from a column that has repeated entries, with no duplicates.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell to the side of the data.
2. Type =UNIQUE(range) and press Enter.

WHAT THE RESULT LOOKS LIKE

A spilled list of only the distinct values appears, even though the source range contains repeats.

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8. Use LEFT, RIGHT, and MID

Your Task: Pull a state code from the start, a ZIP suffix from the end, and the middle characters of an ID from one text cell.

STEP-BY-STEP INSTRUCTIONS

1. Type =LEFT(A2,2) to pull the first two characters.
2. Type =RIGHT(A2,4) to pull the last four characters.
3. Type =MID(A2,3,5) to pull five characters starting at position 3.

WHAT THE RESULT LOOKS LIKE

Each formula returns just the requested slice of the original text string, unaffected by the other formulas.

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9. Use UPPER, LOWER, and LEN

Your Task: Standardize a name to all capitals, all lowercase, and find out how many characters it contains.

STEP-BY-STEP INSTRUCTIONS

1. Type =UPPER(A2) in one cell.
2. Type =LOWER(A2) in another cell.
3. Type =LEN(A2) in a third cell.

WHAT THE RESULT LOOKS LIKE

The formulas return the all-caps version, the all-lowercase version, and the character count of the original text.

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10. Use CONCAT and TEXTJOIN

Your Task: Combine a first and last name into one full-name cell, then join a list of items with commas between them.

STEP-BY-STEP INSTRUCTIONS

1. Type =CONCAT(A2," ",B2) to join a first and last name with a space between.
2. Type =TEXTJOIN(", ",TRUE,range) to join several cells with commas, skipping any blanks.

WHAT THE RESULT LOOKS LIKE

CONCAT returns the combined full name; TEXTJOIN returns a single comma-separated string with blanks automatically skipped.

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11. Use a Nested IF

Your Task: Assign a letter grade (A/B/C/F) based on a score, using more than one condition in a single formula.

STEP-BY-STEP INSTRUCTIONS

1. Type =IF(A2>=90,"A",IF(A2>=80,"B",IF(A2>=70,"C","F"))) in a blank cell.
2. Press Enter, then fill the formula down for the remaining rows.

WHAT THE RESULT LOOKS LIKE

Each row now shows the correct letter grade for its score, based on whichever condition it satisfies first.

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12. Use COUNTIF

Your Task: Count how many rows in a column match a specific condition, such as a particular department name.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell.
2. Type =COUNTIF(range,"Sales") (or another value/condition) and press Enter.

WHAT THE RESULT LOOKS LIKE

The formula returns the number of cells in the range that match the given criterion.

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13. Use SUMIF

Your Task: Total only the amounts that belong to one category, ignoring every other row.

STEP-BY-STEP INSTRUCTIONS

1. Click a blank cell.
2. Type =SUMIF(criteria_range,"Sales",sum_range) and press Enter.

WHAT THE RESULT LOOKS LIKE

The formula returns the sum of only the rows whose criteria cell matches “Sales,” ignoring the rest.

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14. Use TODAY and NOW

Your Task: Stamp a cell with today's date and another with the current date and time.

STEP-BY-STEP INSTRUCTIONS

1. Type =TODAY() in a blank cell and press Enter.
2. Type =NOW() in another blank cell and press Enter.

WHAT THE RESULT LOOKS LIKE

TODAY shows the current date; NOW shows the current date and time, both updating automatically when the workbook recalculates.

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15. Use YEAR, MONTH, and DAY

Your Task: Break a full date in one cell out into separate year, month, and day numbers.

STEP-BY-STEP INSTRUCTIONS

1. Type =YEAR(A2) in a blank cell.
2. Type =MONTH(A2) in another cell.
3. Type =DAY(A2) in a third cell.

WHAT THE RESULT LOOKS LIKE

Each formula extracts just the year, month, or day number from the original date cell.

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16. Use VLOOKUP

Your Task: Look up an employee's department by their ID number from a separate lookup table.

STEP-BY-STEP INSTRUCTIONS

1 Click a blank cell.
2 Type =VLOOKUP(lookup_value,table_range,column_number,FALSE) and press Enter.

WHAT THE RESULT LOOKS LIKE

The formula returns the matching value from the specified column of the lookup table, or #N/A if no match exists.

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17. Use XLOOKUP

Your Task: Repeat the same department lookup using the newer, more flexible XLOOKUP function.

STEP-BY-STEP INSTRUCTIONS

1 Click a blank cell.
2 Type =XLOOKUP(lookup_value,lookup_array,return_array) and press Enter.

WHAT THE RESULT LOOKS LIKE

The formula returns the matching value directly, without needing a column number, and can look left or right of the lookup column.

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18. Use IFERROR

Your Task: Wrap a lookup formula so it shows a friendly message instead of an error when no match is found.

STEP-BY-STEP INSTRUCTIONS

1 Click the cell containing a VLOOKUP or XLOOKUP formula.
2 Wrap it as =IFERROR(original_formula,"Not found") and press Enter.

WHAT THE RESULT LOOKS LIKE

Rows with a valid match show the looked-up value as before; rows with no match now show “Not found” instead of #N/A.

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